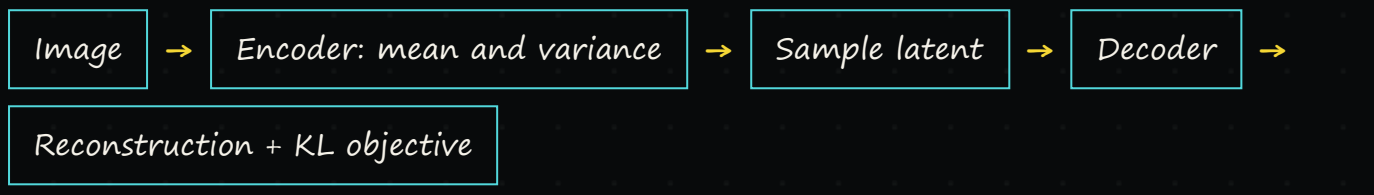
Variational Autoencoders

Turn compression into generation by learning a regularized distribution over latent variables rather than isolated deterministic codes.

Original study notes based on David Foster, Second edition, 2023. Book pp. 59–94. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.



Detailed Notes

Autoencoders and the Fashion-MNIST project

Source: book pp. 59–63 / PDF p. 87

An autoencoder maps an input image through an encoder into a low-dimensional code, then reconstructs it with a decoder. Fashion-MNIST gives a small grayscale setting for understanding the architecture and inspecting latent organization.

The target is the input itself, not the clothing class. Classes can be used after training to visualize whether codes have meaningful structure. Normalize pixels consistently and preserve a held-out reconstruction set. A narrow bottleneck encourages useful compression, but a sufficiently flexible autoencoder can still learn an unhelpful lookup-like mapping.

Encoder, decoder, and reconstruction

Source: book pp. 64–69 / PDF p. 92

The encoder progressively reduces spatial dimensions with convolutions and projects to a latent vector. The decoder expands the vector into spatial feature maps and reconstructs an image, commonly with transposed convolution or upsampling plus convolution.

Join encoder and decoder into one differentiable model and train with reconstruction loss, such as mean squared error or an appropriate likelihood-based loss. Track output size/range and the reduction convention used for summing pixels. Shape mistakes and output activations inconsistent with normalization can prevent sensible learning.

Why plain autoencoders are awkward generators

Source: book pp. 70–74 / PDF p. 98

Reconstruction performance does not tell us where random latent points should be sampled. A deterministic encoder can scatter observations into clusters with large empty gaps. Sampling arbitrary coordinates in those gaps can produce invalid images, and interpolation can traverse unsupported regions.

A generative model needs a prior with a known sampling procedure and a decoder trained to behave usefully across that prior. The VAE adds stochastic latent inference and regularization to encourage a smoother, more usable space.

VAE encoder and the reparameterization trick

Source: book pp. 75–79 / PDF p. 103

The encoder predicts a mean and log variance for a diagonal Gaussian $q(z|x)$, rather than a single latent point. Draw independent standard Gaussian noise and construct the latent sample through a differentiable transform.

> $z = \mu(x) + exp(0.5 \times log\ \sigma^2(x)) \odot \varepsilon$
> $\varepsilon \sim N(0,I)$

Backpropagation then passes through mean/scale while randomness resides in ε. Predicting log variance avoids directly constraining a network output to positive values. Check the distinction between standard deviation and variance; using exp(log variance) as the scale is incorrect.

Loss function: reconstruction and KL divergence

Source: book pp. 80–83 / PDF p. 108

The objective balances accurate reconstruction with matching the approximate posterior to a chosen prior, usually a standard Gaussian. It corresponds to maximizing an evidence lower bound (ELBO), not evaluating exact marginal likelihood.

> $Loss = E_{q(z|x)}[-log\ p(x|z)] + KL(q(z|x)\ ||\ N(0,I))$
> $Gaussian\ KL = -0.5\Sigma(1 + log\ \sigma^2 - \mu^2 - \sigma^2)$

Loss scaling matters: summing reconstruction over pixels versus averaging changes its balance with KL. Too much regularization can erase information; too little leaves poor prior sampling. A low total training loss is not sufficient evidence of good new-image generation.

Training and analyzing the latent space

Source: book pp. 84–89 / PDF p. 112

Inspect reconstructions, sampled prior points, latent-coordinate distributions, and interpolations. The CelebA extension uses richer face images and a larger convolutional architecture, showing how model capacity and latent dimension affect results.

Distinguish stochastic variability on repeated encodings from deliberate semantic edits. Smooth latent organization is encouraged, not guaranteed to disentangle every concept. Pixel-level reconstruction objectives and the probabilistic decoder can yield softer/blurry outputs; this is a trade-off rather than a universal VAE defect.

New faces, latent arithmetic, and morphing

Source: book pp. 90–94 / PDF p. 118

Generate by sampling from the prior and decoding, without requiring an input image. Estimate semantic directions by comparing latent representations of examples with and without an attribute, then adjust another code along that direction.

> Edited code = original code + strength × attribute direction
> Interpolation $z(t) = (1-t)z_A + tz_B$

Decode a sequence of interpolated points to visualize a morph. Attribute correlations can cause several features to change together. Edited faces are model outputs, not evidence that the representation has captured independent causal properties.

Revision Check

1. Why can a good autoencoder reconstruct well but sample badly?
2. What role does the KL term play?
3. Why is the stochastic sample reparameterized?
4. How does the reconstruction reduction affect training?
5. What can go wrong with attribute arithmetic?

VAEs make latent sampling explicit, but useful generation depends on the balance between information preservation and distributional regularity.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.